Session 7 — Mini-challenge (slide 29): unfreeze `layer3` as well?

Computer Vision & Speech Recognition · Joshua Cole · MILK10k skin-lesion classification (3 classes) · seed 0, 1, 2

Repository: https://github.com/jcole-jpg/Computer-Vision-and-Speech-Recognition

Code that completes the exercise: `notebooks/07_cnn.ipynb` (section 5, executed, with outputs) · `src/milk10k/cnn.py` (`resnet18(trainable=...)` , `fit_cnn` , `count_parameters`) · `scripts/build_session7_notebook.py` (generates and runs the notebook) · `tests/test_cnn.py` (tests) · results: `artifacts/session07/mini_challenge.csv` , `artifacts/session07/runs/` (per-epoch history of every run).

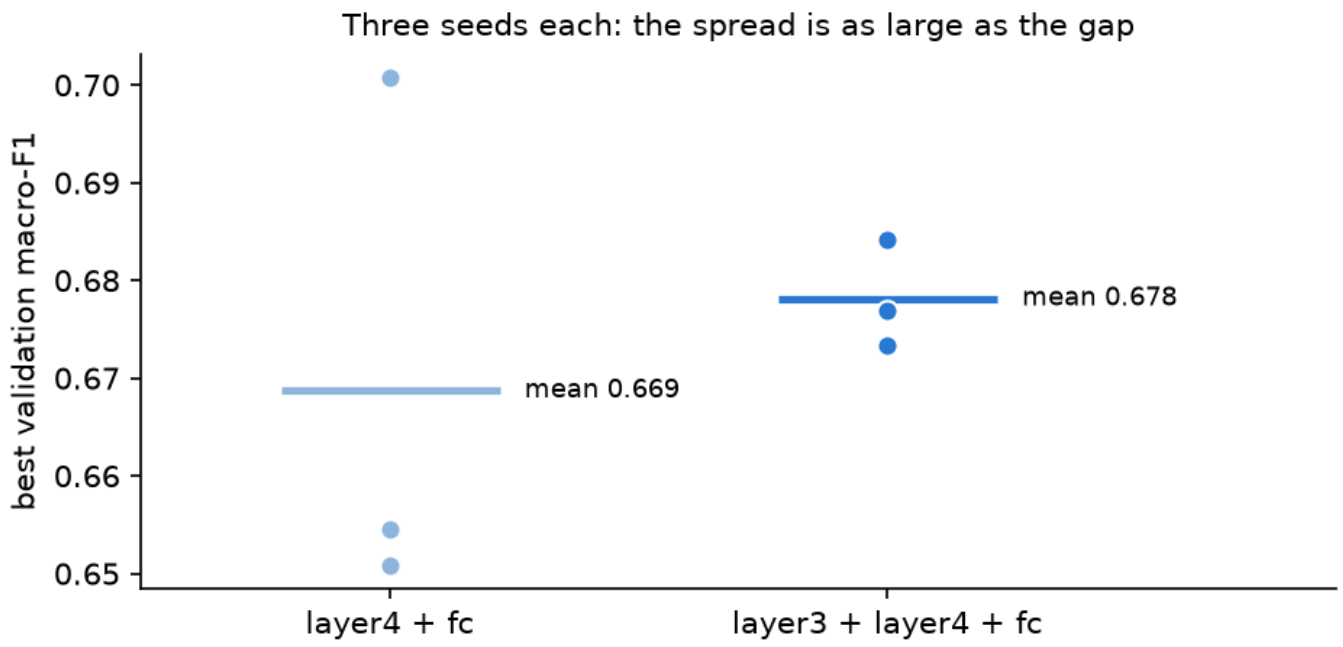
The exercise

Pretrained ImageNet ResNet-18 with a new `fc = Linear(512, 3)` . Compare **`layer4 + fc`** trainable (the usual default) against **`layer3 + layer4 + fc`** trainable. Same data (3,742 / 749 / 749 lesion-level splits), same augmentation, same 10 epochs, class-weighted cross-entropy, Adam with the new head at 1e-3 and the unfrozen pretrained layers at 1e-4. Measured: trainable parameters, seconds per epoch (median over the epochs, so the GPU warm-up of epoch 1 does not count; Apple M2 Pro), best **validation** macro-F1. The test split is not used for this decision.

Results

unfrozen	trainable parameters	seconds / epoch	best val macro-F1, seed 0	mean of 3 seeds	std
`layer4 + fc`	8,395,267	10.9	0.654	0.669	0.023
`layer3 + layer4 + fc`	10,494,979	14.0	0.684	0.678	0.004
difference	+2,099,712 (+25%)	+3.1 (+28%)	+0.030	+0.009	

Seeds 0 / 1 / 2, best validation macro-F1: `layer4 + fc` 0.654 / 0.651 / 0.701; `layer3 + layer4 + fc` 0.684 / 0.677 / 0.673.



Is it worth it?

No, not on this evidence. Unfreezing `layer3` adds 2.1 million trainable weights (+25 %) and 3.1 s per epoch (+28 %), and buys +0.009 validation macro-F1 on average, while `layer4 + fc` alone moves by 0.023 from seed to seed. Seed 0 on its own would have said +0.030; seed 2 puts `layer4 + fc` ahead (0.701 against 0.673). Only 17 Indeterminate lesions are in the validation split, so one lesion moves the class F1 a lot, which is why a single run cannot settle a difference this small. The one consistent effect is a tighter spread with `layer3` unfrozen (0.004 against 0.023), which three seeds can only hint at. The extra time is small in absolute terms (about 30 s over ten epochs), so the verdict is "no evidence it helps", not "it hurts": keep `layer4 + fc` as the default partial fine-tune.

Test split: closed for this decision. The choice above was made on validation only. (The Session 7 scoreboard scores every network once on test, seed 0: `layer4 + fc` and `layer3 + layer4 + fc` both get 0.577, so nothing changes.)